

Name: Sagi Sahasra Srihasa
Roll : 2520030008

Step 1:

```
1  /* Name: Sagi Sahasra Srihasa
2     Roll: 2520030008
3  */
4  CREATE DATABASE Bank_ACID_DB;
5  USE Bank_ACID_DB;
6
7  CREATE TABLE Customer (
8      Customer_ID INT PRIMARY KEY,
9      Customer_Name VARCHAR(100) NOT NULL,
10     Phone VARCHAR(15),
11     City VARCHAR(50)
12 ) ENGINE=InnoDB;
13
14 CREATE TABLE Account (
15     Account_No INT PRIMARY KEY,
16     Customer_ID INT NOT NULL,
17     Account_Type VARCHAR(20),
18     Balance DECIMAL(12,2) NOT NULL,
19     Branch VARCHAR(50),
20     FOREIGN KEY (Customer_ID)
21         REFERENCES Customer(Customer_ID)
22 ) ENGINE=InnoDB;
```

✓	49	13:05:37	CREATE DATABASE Bank_ACID_DB	1 row(s) affected
✓	50	13:05:41	USE Bank_ACID_DB	0 row(s) affected
✓	51	13:05:44	CREATE TABLE Customer (Customer_ID INT ...	0 row(s) affected
✓	52	13:05:49	CREATE TABLE Account (Account_No INT P...	0 row(s) affected

Step 2:

```
INSERT INTO Customer
(Customer_ID, Customer_Name, Phone, City)
VALUES
(101, 'Sahasra', '9876543210', 'Hyderabad'),
(102, 'Priya Sharma', '9876543211', 'Vijayawada'),
(103, 'Arjun Reddy', '9876543212', 'Bangalore'),
(104, 'Sneha Rao', '9876543213', 'Chennai'),
(105, 'Kiran Kumar', '9876543214', 'Hyderabad');

INSERT INTO Account
(Account_No, Customer_ID, Account_Type, Balance, Branch)
VALUES
(10001, 101, 'Savings', 50000.00, 'Hyderabad'),
(10002, 102, 'Savings', 75000.00, 'Vijayawada'),
(10003, 103, 'Current', 120000.00, 'Bangalore'),
(10004, 104, 'Savings', 45000.00, 'Chennai'),
(10005, 105, 'Current', 90000.00, 'Hyderabad');
```

✓	53	13:08:00	INSERT INTO Customer (Customer_ID, Customer...	5 row(s) affected Records: 5 Duplicates: 0 Wami...	0.010
✓	54	13:08:02	INSERT INTO Account (Account_No, Customer_...	5 row(s) affected Records: 5 Duplicates: 0 Wami...	0.010

Step 3:

42 • **SELECT * FROM Account;**

43

Result Grid

Filter Rows:

Edit:   

Export/Import:  

W

	Account_No	Customer_ID	Account_Type	Balance	Branch
▶	10001	101	Savings	50000.00	Hyderabad
	10002	102	Savings	75000.00	Vijayawada
	10003	103	Current	120000.00	Bangalore
	10004	104	Savings	45000.00	Chennai
	10005	105	Current	90000.00	Hyderabad

Step 4:

```
-- ACID PROPERTIES
-- 1. ATOMICITY
-- Transfer Rs.10000 from Account 10001 to 10002
```

- `START TRANSACTION;`
- `UPDATE Account`
`SET Balance = Balance - 10000`
`WHERE Account_No = 10001`
`AND Balance >= 10000;`
- `UPDATE Account`
`SET Balance = Balance + 10000`
`WHERE Account_No = 10002;`
- `COMMIT;`

✓	56	13:09:58	START TRANSACTION	0 row(s) affected	0.016 sec
✓	57	13:10:00	UPDATE Account SET Balance = Balance - 100...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
✓	58	13:10:02	UPDATE Account SET Balance = Balance + 100...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
✓	59	13:10:05	COMMIT	0 row(s) affected	0.016 sec

Step 5:

```
62 • SELECT Account_No, Balance
63 FROM Account
64 WHERE Account_No IN (10001, 10002);
65
```

Result Grid		Filter Rows:	Edit:	Export/Import:	
	Account_No	Balance			
▶	10001	40000.00			
	10002	85000.00			
•	NULL	NULL			

Step 6:

```
-- 2. ATOMICITY USING ROLLBACK
```

```
START TRANSACTION;
```

```
UPDATE Account
```

```
SET Balance = Balance - 20000
```

```
WHERE Account_No = 10001
```

```
AND Balance >= 20000;
```

```
UPDATE Account
```

```
SET Balance = Balance + 20000
```

```
WHERE Account_No = 10002;
```

```
ROLLBACK;
```

✓	62	13:11:28	START TRANSACTION	0 row(s) affected	0.000 sec
✓	63	13:11:31	UPDATE Account SET Balance = Balance - 200...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
✓	64	13:11:34	UPDATE Account SET Balance = Balance + 200...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
✓	65	13:11:37	ROLLBACK	0 row(s) affected	0.015 sec

Step 7:

```
82 • SELECT Account_No, Balance
```

```
83 FROM Account
```

```
84 WHERE Account_No IN (10001, 10002);
```

```
85
```

Result Grid	Filter Rows:	Edit:	Export/Import:	W
	Account_No	Balance		
▶	10001	40000.00		
	10002	85000.00		
*	NULL	NULL		

Step 8:

```
87      -- 3. CONSISTENCY
88
89 •    SELECT SUM(Balance) AS Total_Before
90      FROM Account;
91
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Total_Before
380000.00

Step 9:

```
START TRANSACTION;

UPDATE Account
SET Balance = Balance - 10000
WHERE Account_No = 10001
AND Balance >= 10000;

UPDATE Account
SET Balance = Balance + 10000
WHERE Account_No = 10002;

COMMIT;
```

✓	68	13:13:25	START TRANSACTION	0 row(s) affected	0.000
✓	69	13:13:27	UPDATE Account SET Balance = Balance - 100...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000
✓	70	13:13:29	UPDATE Account SET Balance = Balance + 100...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000
✓	71	13:13:31	COMMIT	0 row(s) affected	0.000

Step 10:

```
105 •    SELECT SUM(Balance) AS Total_After
106      FROM Account;
107
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Total_After
380000.00

Step 11:

```
109  -- 4. ISOLATION
110
111  •  START TRANSACTION;
112
113  •  UPDATE Account
114      SET Balance = Balance + 5000
115      WHERE Account_No = 10001;
116
117  •  SELECT Balance
118      FROM Account
119      WHERE Account_No = 10001;
120
121  •  COMMIT;
122
123
```

Result Grid | | Filter Rows: | Export: | Wrap Cell Content:

	Balance
▶	35000.00

Account 8 x

Output

Action Output ▼

	#	Time	Action	Message
✓	73	13:14:33	START TRANSACTION	0 row(s) affected
✓	74	13:14:36	UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows matched: 1
✓	75	13:14:43	SELECT Balance FROM Account WHERE Acco...	1 row(s) returned
✓	76	13:14:47	COMMIT	0 row(s) affected
✓	77	13:14:51	SELECT Balance FROM Account WHERE Acco...	1 row(s) returned

Step 12:

```
124      -- 5. DURABILITY
125
126 •    START TRANSACTION;
127
128 •    UPDATE Account
129      SET Balance = Balance + 5000
130      WHERE Account_No = 10001;
131
132 •    COMMIT;
133
134 •    SELECT Account_No, Balance
135      FROM Account
136      WHERE Account_No = 10001;
137
```

Result Grid		Filter Rows:	Edit:
	Account_No	Balance	
▶	10001	40000.00	
✱	NULL	NULL	

✓	79	13:16:31	START TRANSACTION	0 row(s) affected	0.
✓	80	13:16:35	UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.
✓	81	13:16:38	COMMIT	0 row(s) affected	0.
✓	82	13:16:40	SELECT Account_No, Balance FROM Account ...	1 row(s) returned	0.

Step 13:

```
140 -- QUESTION 1
141 -- Deposit Rs.5000 into Account 10001 and COMMIT
142
143 • START TRANSACTION;
144
145 • UPDATE Account
146 SET Balance = Balance + 5000
147 WHERE Account_No = 10001;
148
149 • COMMIT;
150
151 • SELECT Account_No, Balance
152 FROM Account
153 WHERE Account_No = 10001;
154
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

	Account_No	Balance
▶	10001	45000.00
*	NULL	NULL

Account 10 x Apply

Output :.....

Action Output ▾

	#	Time	Action	Message	Duration / Fetch
✓	82	13:16:40	SELECT Account_No, Balance FROM Account ...	1 row(s) returned	0.000 sec / 0.00
✓	83	13:17:41	START TRANSACTION	0 row(s) affected	0.000 sec
✓	84	13:17:45	UPDATE Account SET Balance = Balance + 500...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
✓	85	13:17:47	COMMIT	0 row(s) affected	0.000 sec
✓	86	13:17:50	SELECT Account_No, Balance FROM Account ...	1 row(s) returned	0.000 sec / 0.00

Step 14:

```
156 -- QUESTION 2
157 -- Withdraw Rs.3000 from Account 10002 and COMMIT
158 • START TRANSACTION;
159 • UPDATE Account
160   SET Balance = Balance - 3000
161   WHERE Account_No = 10002
162   AND Balance >= 3000;
163
164 • COMMIT;
165
166 • SELECT Account_No, Balance
167   FROM Account
168   WHERE Account_No = 10002;
169
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	Account_No	Balance
▶	10002	92000.00
*	NULL	NULL

Account 11 x Apply Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 86	13:17:50	SELECT Account_No, Balance FROM Account ...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 87	13:18:19	START TRANSACTION	0 row(s) affected	0.000 sec
✓ 88	13:18:22	UPDATE Account SET Balance = Balance - 300...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
✓ 89	13:18:24	COMMIT	0 row(s) affected	0.031 sec
✓ 90	13:18:29	SELECT Account_No, Balance FROM Account ...	1 row(s) returned	0.000 sec / 0.000 sec

Step 15:

```
171  -- QUESTION 3
172  -- Withdraw Rs.10000 and ROLLBACK
173  •  START TRANSACTION;
174  •  UPDATE Account
175     SET Balance = Balance - 10000
176     WHERE Account_No = 10001
177        AND Balance >= 10000;
178
179  •  ROLLBACK;
180
181  •  SELECT Account_No, Balance
182     FROM Account
183     WHERE Account No = 10001;
```

Result Grid

	Account_No	Balance
▶	10001	45000.00
*	NULL	NULL

Account 12 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 90	13:18:29	SELECT Account_No, Balance FROM Account ...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 91	13:19:01	START TRANSACTION	0 row(s) affected	0.000 sec
✓ 92	13:19:03	UPDATE Account SET Balance = Balance - 100...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
✓ 93	13:19:05	ROLLBACK	0 row(s) affected	0.015 sec
✓ 94	13:19:08	SELECT Account_No, Balance FROM Account ...	1 row(s) returned	0.000 sec / 0.000 sec

Step 16:

```
187  -- Transfer Rs.15000 from Account 10001 to Account 10002
188
189  •  START TRANSACTION;
190  •  UPDATE Account
191     SET Balance = Balance - 15000
192     WHERE Account_No = 10001
193     AND Balance >= 15000;
194  •  UPDATE Account
195     SET Balance = Balance + 15000
196     WHERE Account_No = 10002;
197
198  •  COMMIT;
199  •  SELECT Account_No, Balance
200     FROM Account
201     WHERE Account No IN (10001, 10002);
```

Result Grid

Account_No	Balance
10001	30000.00
10002	107000.00
NULL	NULL

Account 13 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 95	13:19:50	START TRANSACTION	0 row(s) affected	0.000 sec
✓ 96	13:19:53	UPDATE Account SET Balance = Balance - 150...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
✓ 97	13:19:55	UPDATE Account SET Balance = Balance + 150...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
✓ 98	13:19:57	COMMIT	0 row(s) affected	0.000 sec
✓ 99	13:19:59	SELECT Account_No, Balance FROM Account ...	2 row(s) returned	0.000 sec / 0.000 sec

Step 17:

```
-- QUESTION 5
-- Transfer Rs.20000 from Account 10001 to Account 10003
-- using SAVEPOINT
```

- `START TRANSACTION;`
- `UPDATE Account`
`SET Balance = Balance - 20000`
`WHERE Account_No = 10001`
`AND Balance >= 20000;`
- `SAVEPOINT AfterDebit;`
- `UPDATE Account`
`SET Balance = Balance + 20000`
`WHERE Account_No = 10003;`
- `COMMIT;`

✓	100	13:21:00	START TRANSACTION	0 row(s) affected	0.000 sec
✓	101	13:21:02	UPDATE Account SET Balance = Balance - 200...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
✓	102	13:21:05	SAVEPOINT AfterDebit	0 row(s) affected	0.000 sec
✓	103	13:21:07	UPDATE Account SET Balance = Balance + 200...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
✓	104	13:21:09	COMMIT	0 row(s) affected	0.000 sec

Step 18:

```
223 • SELECT Account_No, Balance
224 FROM Account
225 WHERE Account_No IN (10001, 10003);
```

Result Grid

	Account_No	Balance
▶	10001	10000.00
	10003	140000.00
*	NULL	NULL

Filter Rows: Edit: Export/Import: Wrap Cell Content: